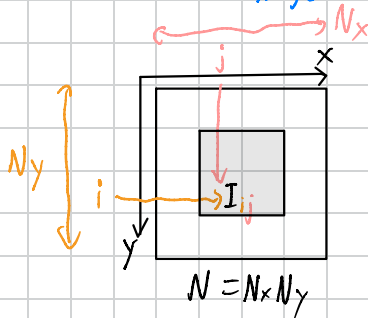


Quantum Hadamard Edge detection (QHED)

- ① Definition of edge
- ② Classical edge detection
- ③ QHED

① Definition

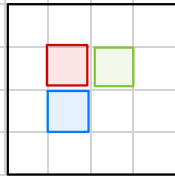
(i) Definition of Image



- For B&W image, each pixel is represented by I_{ij}
black $\longleftrightarrow$ white

Usually, $I_{ij} \in [0, 255]$

$256 = 2^8$, known as 8-bit colour
 $\Rightarrow$ representing 256 type of colour



- For RGB image each pixel is represented

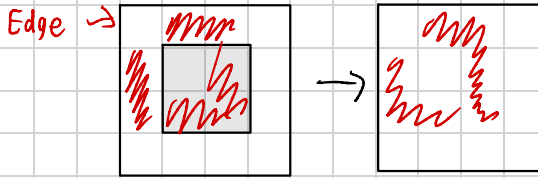
by vector $\vec{v}$
$$\vec{v} = \begin{pmatrix} R_{ij} \\ G_{ij} \\ B_{ij} \end{pmatrix}$$

$R_{ij}, G_{ij}, B_{ij} \in [0, 255]$

We can convert RGB $\rightarrow$ B&W using the following function

$$I_{ij} = 0.299R + 0.587G + 0.114B$$

(ii) Definition of Edge



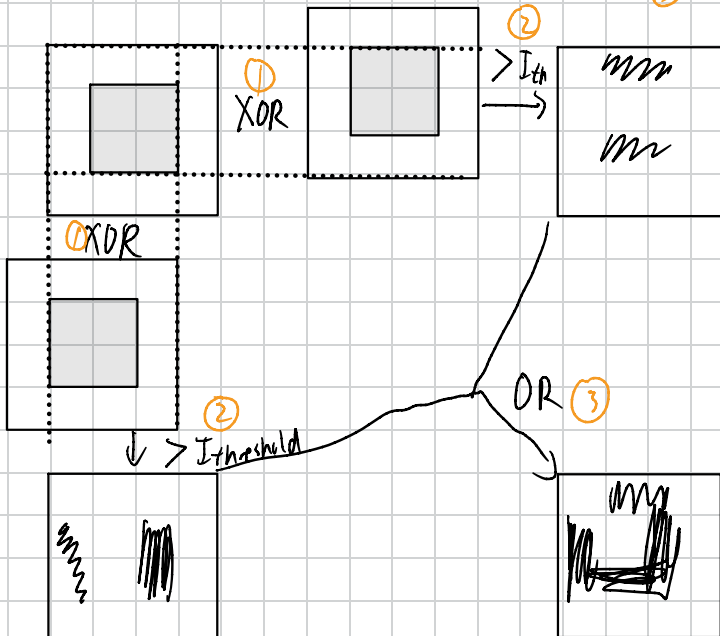
We say pixel (i, j) is edge iff

$$|I_{i+1,j} - I_{i,j}| \geq I_{\text{threshold}} \\ \text{OR } |I_{i,j+1} - I_{i,j}| \geq I_{\text{threshold}}$$

Where $I_{\text{threshold}}$ is a hyperparameter for tuning

② Classical edge detection

Edge can be detected by (XOR gate on 2' inputs)
 performing ① elementwise absolute difference
 in vertical and horizontal direction
 ② Elementwise comparison
 ③ ORing them



Binary Image edge & Details of B&W edge can be found in supplementary

Problem of classical approach:

$$\text{Let } n_x = \lceil \log_2 N_x \rceil$$

$$n_y = \lceil \log_2 N_y \rceil$$

$$\text{Then } \# \text{FLOP} = \text{XOR} + >0 + \text{OR}$$

$$O(N_x N_y) \quad O(N_x N_y) \quad O(N_x N_y)$$

$$= O(N_x N_y)$$

$$= O(2^{n_x + n_y})$$

$\Rightarrow$ exponential time

② QHED:

If we neglect the time for encoding image & measurement

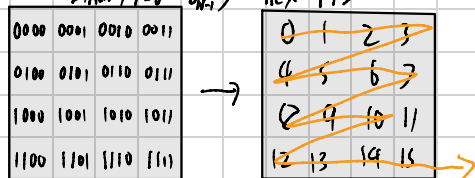
(i) Quantum probability image encoding (QPIE)

we can represent intensity as probability

$$C_{ij} = \frac{I_{ij}}{\sum_{i,j} I_{ij}} \quad (\text{normalisation})$$

Each pixel $\equiv$ each basis vector

For simplicity, assume $N_x, N_y = \text{even}$,
binary $|b_0 \dots b_{n_x}\rangle$ hex $|i\rangle$



Using row-major order, we can unroll image into 1D-array
 $C_{ij} \rightarrow C_i$

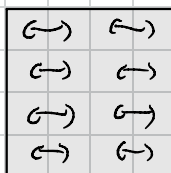
$$| \text{img} \rangle = \sum_i C_i |i\rangle \quad (\text{State preparation})$$

• It can be proven that arbitrary state can be prepared using only (NOT & R(0, ϕ))

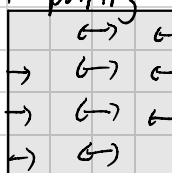
• It can be proven that such encoding have runtime $O(n^2)$

(ii) QHED:

Define even & odd pairing as even pairing
odd & even pairing as odd pairing



even pairing



odd pairing

Goal: Find their absolute difference of even & odd pairing

If we can do this,
then we achieve

horizontal scanning ✓
Vertical scanning = horizontal scan(image^T)

Method ①

• For

even-pairing:

↔ ↔

0000	0001	0010	0011
0100	0101	0110	0111
1000	1001	1010	1011
1100	1101	1110	1111

One notice that only least significant bit is different

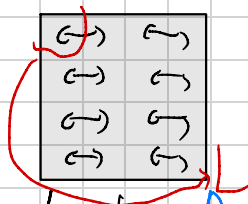
By the magic of H-gate

$$I_{N-1} \otimes H_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ & & 1 & 1 \\ & & 1 & -1 \\ & & & & \ddots & \ddots \end{pmatrix}$$

$$\begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_{N-1} \end{pmatrix} = \begin{pmatrix} c_0 + c_1 \\ c_0 - c_1 \\ c_2 + c_3 \\ c_2 - c_3 \\ \vdots \end{pmatrix}$$

✓ Difference in O(1)

• For odd-pairing: We just need to append first pixel to the last



This can be done by **Decrement gate**

$$\begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix} \begin{pmatrix} c_0 \\ \vdots \\ c_{N-1} \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_0 \end{pmatrix}$$

Then repeat last procedure

Method ②

Auxiliary qubit

One realise even pairing & Odd pairing can be done at the same time by adding 1 extra qubit

And Apply **$|1\rangle_{\text{mg}} \otimes |0\rangle$**
H-gate to it

$$\frac{|1\rangle_{\text{mg}} \otimes |0\rangle + |1\rangle}{\sqrt{2}} = \begin{pmatrix} c_0 \\ c_0 \\ c_1 \\ c_1 \\ \vdots \\ c_{N-1} \\ c_{N-1} \end{pmatrix}$$

Apply **Decrement gate** Then Hgate

$$I_{N+1} \otimes H_0 \begin{pmatrix} c_0 \\ c_1 \\ c_1 \\ c_2 \\ \vdots \\ c_{N-1} \\ c_{N-1} \\ c_0 \end{pmatrix} = \begin{pmatrix} c_0 + c_1 \\ c_0 - c_1 \\ c_1 + c_2 \\ c_1 - c_2 \\ \vdots \\ c_{N-1} + c_0 \\ c_{N-1} - c_0 \end{pmatrix}$$

✓ Diff. in O(1)